STM32 CubeMX

1. Description

1.1. Project

Project Name	SSP_523
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	05/05/2026

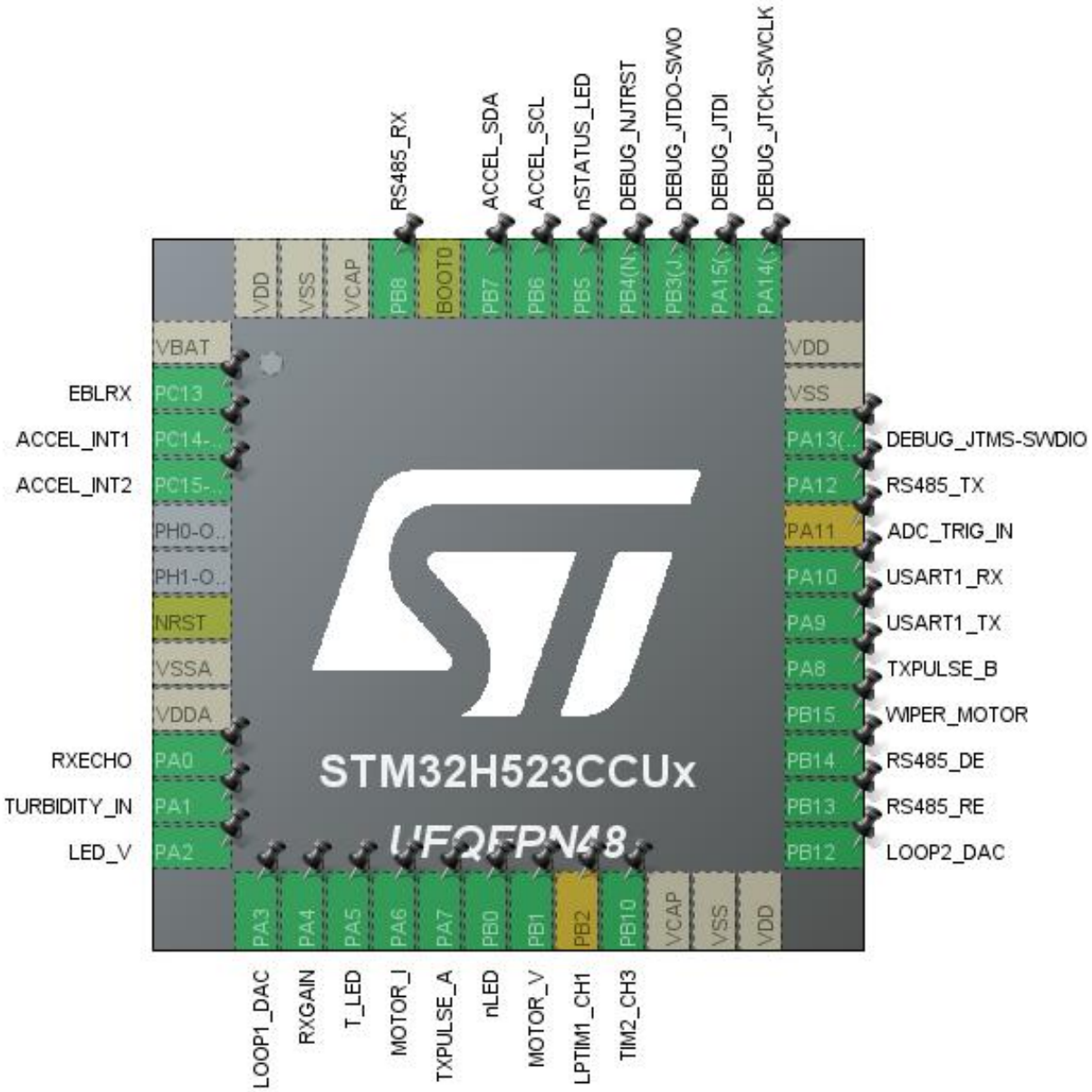
1.2. MCU

MCU Series	STM32H5
MCU Line	STM32H5x3
MCU name	STM32H523CCUx
MCU Package	UFQFPN48
MCU Pin number	48

1.3. Core(s) information

Core(s)	ARM Cortex-M33

2. Pinout Configuration



3. Pins Configuration

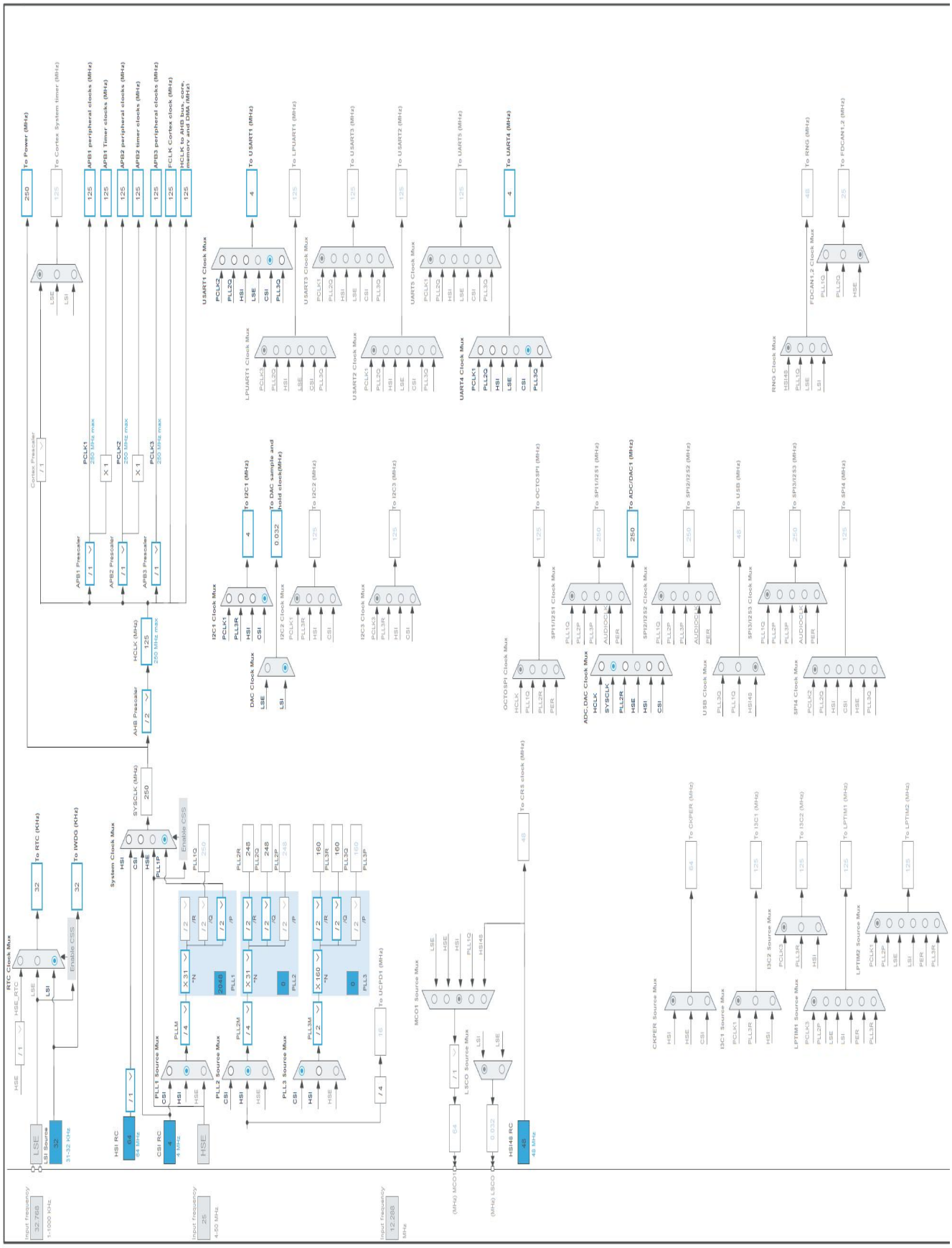
Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13 *	I/O	GPIO_Output	EBLRX
3	PC14-OSC32_IN(OSC32_IN) *	I/O	GPIO_Input	ACCEL_INT1
4	PC15-OSC32_OUT(OSC32_OUT) *	I/O	GPIO_Input	ACCEL_INT2
7	NRST	Reset		
8	VSSA	Power		
9	VDDA	Power		
10	PA0	I/O	ADC1_INP0, ADC2_INP0	RXECHO
11	PA1	I/O	ADC1_INP1	TURBIDITY_IN
12	PA2	I/O	ADC1_INP14	LED_V
13	PA3	I/O	TIM15_CH2	LOOP1_DAC
14	PA4	I/O	DAC1_OUT1	RXGAIN
15	PA5 *	I/O	GPIO_Output	T_LED
16	PA6	I/O	ADC1_INP3	MOTOR_I
17	PA7	I/O	TIM1_CH1N	TXPULSE_A
18	PB0 *	I/O	GPIO_Output	nLED
19	PB1	I/O	ADC1_INP5	MOTOR_V
20	PB2 **	I/O	LPTIM1_CH1	
21	PB10	I/O	TIM2_CH3	
22	VCAP	Power		
23	VSS	Power		
24	VDD	Power		
25	PB12	I/O	TIM8_CH3	LOOP2_DAC
26	PB13 *	I/O	GPIO_Output	RS485_RE
27	PB14	I/O	UART4_DE	RS485_DE
28	PB15 *	I/O	GPIO_Output	WIPER_MOTOR
29	PA8	I/O	TIM1_CH1	TXPULSE_B
30	PA9	I/O	USART1_TX	
31	PA10	I/O	USART1_RX	
32	PA11 **	I/O	ADC1_EXTI11, ADC2_EXTI11	ADC_TRIG_IN
33	PA12	I/O	UART4_TX	RS485_TX
34	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
35	VSS	Power		
36	VDD	Power		
37	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
38	PA15(JTDI)	I/O	DEBUG_JTDI	
39	PB3(JTDO/TRACESWO)	I/O	DEBUG_JTDO-SWO	
40	PB4(NJTRST)	I/O	DEBUG_NJTRST	
41	PB5 *	I/O	GPIO_Output	nSTATUS_LED
42	PB6	I/O	I2C1_SCL	ACCEL_SCL
43	PB7	I/O	I2C1_SDA	ACCEL_SDA
44	BOOT0	Boot		
45	PB8	I/O	UART4_RX	RS485_RX
46	VCAP	Power		
47	VSS	Power		
48	VDD	Power		

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H5
Line	STM32H5x3
MCU	STM32H523CCUx
Datasheet	DS00000_Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

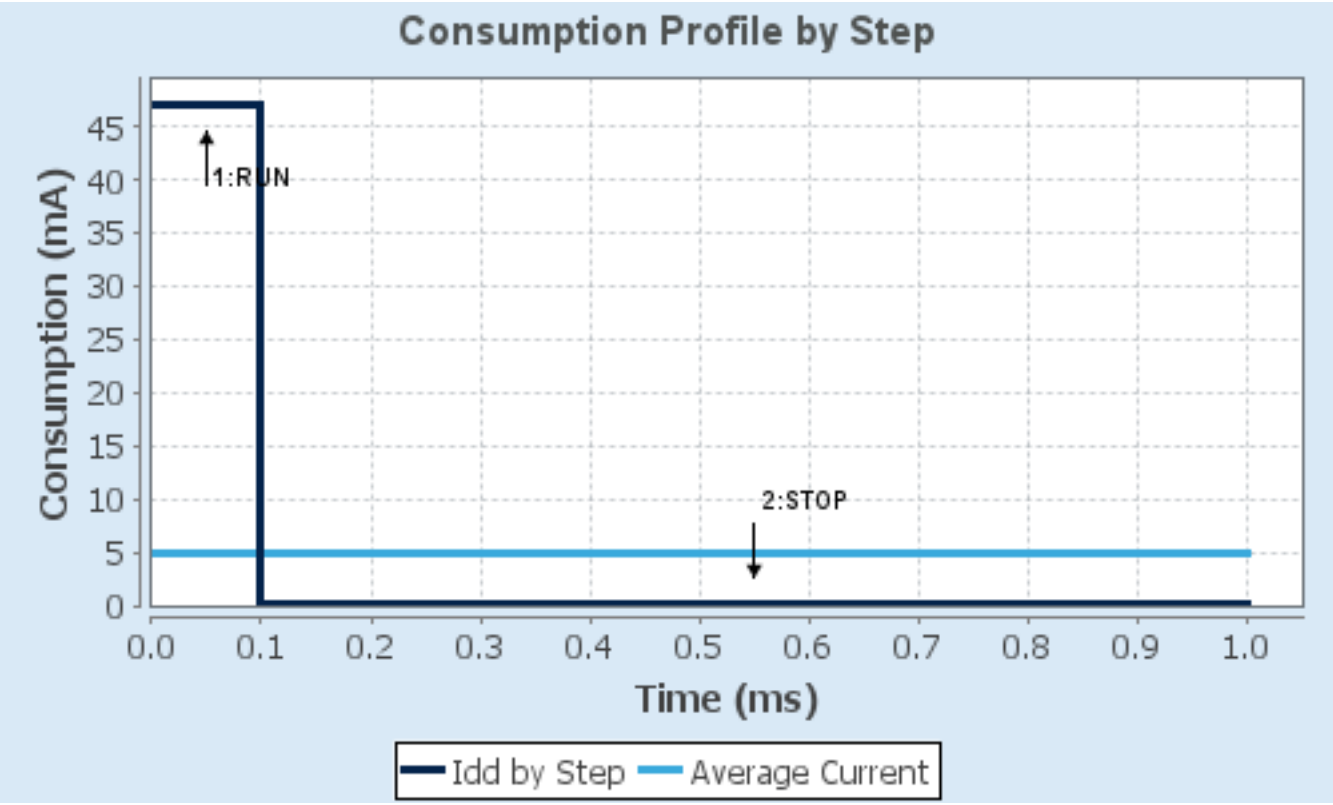
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5
Fetch Type	FLASH_ON/Cache2Ways_ALL_RAM_RETENTION	FLASH_PwrDwnStop_OFF
CPU Frequency	250 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals	ADC1 DAC1 DTS FLASH GPDMA1 GPDMA2 GPIOA GPIOB GPIOC I2C1 ICACHE LPTIM1 LPUART1 RAMCFG SRAM1 SRAM2 SRAM3 TIM1 TIM2 TIM3 TIM4 TIM5 TIM6 TIM7 TIM8 TIM12 TIM15 UART4 USART1 VREF	
Additional Cons.	0 mA	0 mA
Average Current	47.16 mA	100 µA
Duration	0.1 ms	0.9 ms
DMIPS	375.0	0.0
Ta Max	125	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	4.81 mA
Battery Life	29 days, 10 hours	Average DMIPS	37.5 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
45
40
35
30
25
20
15
10
5
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	SSP_523
Project Folder	C:\Projects_local\01-0433_EchoSmart_git
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H5 V1.5.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x400
Minimum Stack Size	0x800

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	Yes

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ICACHE_Init	ICACHE
4	MX_DCACHE1_Init	DCACHE1
5	MX_FLASH_Init	FLASH
6	MX_TIM1_Init	TIM1
7	MX_TIM2_Init	TIM2
8	MX_TIM7_Init	TIM7
9	MX_TIM8_Init	TIM8
10	MX_TIM12_Init	TIM12
11	MX_TIM15_Init	TIM15

Rank	Function Name	Peripheral Instance Name
12	MX_USART1_UART_Init	USART1
13	MX_UART4_Init	UART4
14	MX_ADC1_Init	ADC1
15	MX_ADC2_Init	ADC2
16	MX_DAC1_Init	DAC1
17	MX_DTS_Init	DTS
18	MX_I2C1_Init	I2C1
19	MX_IWDG_Init	IWDG
20	MX_TIM5_Init	TIM5
21	MX_TIM4_Init	TIM4
22	MX_CRC_Init	CRC
23	MX_RTC_Init	RTC

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN3: IN3 Single-ended

IN5: IN5 Single-ended

mode: IN14

mode: Temperature Sensor Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Scan Conversion Mode	Disabled
Data Alignment	Right alignment
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	**Enable ***
Oversampling Right Shift	**8 bit shift for oversampling ***
Oversampling Ratio	**Oversampling ratio 256x ***
Regular Oversampling Mode	Oversampling Continued Mode
Triggered Regular Oversampling	Single trigger for all oversampled conversions
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Sampling Mode	Normal
Rank	1
Channel	Channel 1
Sampling Time	2.5 Cycles
Offset Number	No offset
Monitored by	None

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.2. ADC2

mode: IN0

3.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Scan Conversion Mode	Disabled
Data Alignment	Right alignment
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	**Overrun data overwritten ***
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Sampling Mode	**Bulb ***
Rank	1
Channel	Channel 0
Sampling Time	2.5 Cycles
Offset Number	No offset
Monitored by	None

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.3. BOOTPATH
mode: Activated

3.4. CRC
mode: Activated
3.4.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.5. DAC1
OUT1 connected to: only external pin
3.5.1. Parameter Settings:

Common DAC Settings:

DAC High Frequency Mode	Disable

DAC Out1 Settings:

Mode selected	Normal Mode
Output Buffer	Enable
DMA double data mode	Disable
Signed Format	Disable
Trigger	None
User Trimming	Factory trimming

3.6. DCACHE1

mode: Activated

3.6.1. Parameter Settings:

Basic Parameters:

DCACHE Read Burst Type	DCACHE READ BURST WRAP

3.7. DEBUG

Debug: JTAG (5 pins)

3.8. DTS

mode: Activated

3.8.1. Parameter Settings:

Basic parameters:

Reference Clock Selection	High speed reference clock (PCLK)
Quick Measurement	Disable the Quick Measurement (Measure with calibration)
Input Trigger Selection	No Hardware trigger detection
Sampling Time	**15 cycles for the sampling time ***
Divider	0
High Threshold	**0x0 ***
Low Threshold	**0x0 ***

3.9. FLASH

mode: Enable

3.9.1. HAL Option Bytes Settings:

Advanced settings:

Activate	false

Product State:

Activate	false

BOR Level:

Activate	false

User Configuration:

Activate	false

User Configuration 2:

Activate	false

Boot Configuration:

Activate NS BOOT LOCK	false

Write sector group Protection:

Activate Bank 1 WRP	false
Activate Bank 2 WRP	false

OTP write protection:

Activate	false

Flash data sectors:

Activate Bank1 EDATA	false
Activate Bank2 EDATA	false

Flash HDP:

Activate Bank1 HDP	false
Activate Bank2 HDP	false

3.9.2. Block-based sector protection:

AREA1:

Activate AREA 1	false

AREA2:

Activate AREA 2	false

AREA3:

Activate AREA 3	false

AREA4:

Activate AREA 4	false

AREA5:

Activate AREA 5	false

AREA6:

Activate AREA 6	false

AREA7:

Activate AREA 7	false

AREA8:

Activate AREA 8	false

3.10. I2C1

I2C: I2C

3.10.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	**Fast Mode ***
I2C Speed Frequency (KHz)	400
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x00000004 ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.11. ICACHE

mode: Memory address remap

3.11.1. Parameter Settings:

Region 0:

Region	Disable

Region 1:

Region	Disable

Region 2:

Region	Disable

Region 3:

Region	Disable

3.12. IWDG

mode: Activated

3.12.1. Parameter Settings:

Watchdog Clocking:

IWDG counter clock prescaler	4
IWDG window value	4095
IWDG down-counter reload value	4095

Watchdog Interrupt:

IWDG Early Wakeup Interrupt	0

3.13. MEMORYMAP
mode: Activated

3.14. PWR
mode: Dead Battery Signals disabled
mode: Power saving mode
mode: Privilege attributes

3.14.1. PWR Privilege :

Privilege PWR:

PWR Privilege	Disable

3.15. RCC

3.15.1. RCC Privilege :

Privilege RCC:

Privilege of RCC Non-Secure Items	Disable

3.15.2. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	5 WS (6 CPU cycle)
Flash Programming Delay	2

RCC Parameters:

HSI Calibration Value	64
CSI Calibration Value	32
HSE Startup Timout Value (ms)	100

LSE Startup Timout Value (ms)	5000
TIM Prescaler Selection	Disabled

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

3.16. RTC

mode: Activate Clock Source

WakeUp: Internal WakeUp

3.16.1. Parameter Settings:

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	127
Synchronous Predivider value	255
Bin Mode	Free running BCD calender mode

Wake UP:

Wake Up Clock	RTCCLK / 16
Wake Up Counter	0
Wake Up Auto Clr	0

3.16.2. RTC Privilege:

Privilege RTC:

RTC full privilege	Disable

Backup register:

Start zone 1	RTC_BKP_DR0
Start Zone 2	RTC_BKP_DR0
start zone 3	RTC_BKP_DR0

Privilege Backup register :

Backup Register PrivZone	Non-privilege

Privilege RTC Feature:

RTC Initialisation	Non-Privilege
RTC Alarm A	Non-Privilege
RTC Alarm B	Non-Privilege
RTC Calibration	Non-Privilege
RTC TimeStamp	Non-Privilege
RTC WakeUpTimer	Non-Privilege

3.17. SYS

Timebase Source: TIM6

3.18. TIM1

Clock Source : Internal Clock

Channel1: PWM Generation CH1 CH1N

3.18.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	**186 ***
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	**22 ***
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off
DeadTime Preload	Disable

Dead Time	**2 ***
Asymmetrical DeadTime	Disable
Falling Dead Time	0

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	**46 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

3.19. TIM2

Clock Source : Internal Clock

Channel3: PWM Generation CH3

3.19.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.20. TIM4

Clock Source : Internal Clock

3.20.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**11 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	**52 ***
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	**Update Event ***

3.21. TIM5

Clock Source : Internal Clock

3.21.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**124 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

3.22. TIM7

mode: Activated

3.22.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)

3.23. TIM8

Clock Source : Internal Clock

Channel3: PWM Generation CH3

3.23.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.24. TIM12

mode: Clock Source

3.24.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

3.25. TIM15

Channel2: PWM Generation CH2

3.25.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0

auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.26. UART4

Mode: Asynchronous

mode: Hardware Flow Control (RS485)

3.26.1. Parameter Settings:

Basic Parameters:

Baud Rate	**9600 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	**FIFO mode enable ***
Txfifo Threshold	**half full configuration ***

Rxfifo Threshold	**half full configuration ***
Polarity	High
Assertion Time	**16 ***
Deassertion Time	**8 ***

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.27. USART1

Mode: Asynchronous

3.27.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable

MSB First	Disable

3.28. THREADX

mode: Core

3.28.1. ThreadX:

Core:

TX_MINIMUM_STACK	200
TX_THREAD_USER_EXTENSION	
TX_DISABLE_STACK_FILLING	Disabled
TX_ENABLE_STACK_CHECKING	Disabled
TX_DISABLE_PREEMPTION_THRESHOLD	Enabled
TX_DISABLE_REDUNDANT_CLEARING	Disabled
TX_DISABLE_NOTIFY_CALLBACKS	Enabled
TX_INLINE_THREAD_RESUME_SUSPEND	Disabled
TX_NOT_INTERRUPTABLE	Disabled
TX_MAX_PRIORITIES	32
TX_TIMER_TICKS_PER_SECOND	100
TX_NO_FILEX_POINTER	Disabled
Enable BASEPRI support	Disabled
TX_DISABLE_ERROR_CHECKING	Disabled

Timer:

TX_TIMER_PROCESS_IN_ISR	Disabled
TX_REACTIVATE_INLINE	Disabled
TX_TIMER_THREAD_STACK_SIZE	1024
TX_TIMER_THREAD_PRIORITY	0

Version:

ThreadX version	6.4.0

ThreadX App Init:

Generate App Init Code	**true ***
Create ThreadX Application Thread	**true ***
ThreadX Application Thread Entry	tx_app_thread_entry
ThreadX Application Thread Name	tx app thread
ThreadX Application Thread Priority	10
ThreadX Application Thread Stack Size	512
Create ThreadX Application Semaphore	false
Create ThreadX Application Mutex	false
Create ThreadX Application MessageQueue	**true ***
ThreadX Application Queue Name	**mcp_cmd_queue ***

ThreadX Application Queue Message Size in words	1
ThreadX Application Number of Messages	**10 ***

Memory Configuration:

ThreadX memory pool size	**61440 ***
Memory Pool Allocation	Use Static Allocation
ThreadX MemPool Name	tx_app_byte_pool

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA0	ADC1_INP0	Analog mode	No pull-up and no pull-down	n/a	RXECHO
	PA1	ADC1_INP1	Analog mode	No pull-up and no pull-down	n/a	TURBIDITY_IN
	PA2	ADC1_INP14	Analog mode	No pull-up and no pull-down	n/a	LED_V
	PA6	ADC1_INP3	Analog mode	No pull-up and no pull-down	n/a	MOTOR_I
	PB1	ADC1_INP5	Analog mode	No pull-up and no pull-down	n/a	MOTOR_V
ADC2	PA0	ADC2_INP0	Analog mode	No pull-up and no pull-down	n/a	RXECHO
DAC1	PA4	DAC1_OUT1	Analog mode	No pull-up and no pull-down	n/a	RXGAIN
DEBUG	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
	PA15(JTDI)	DEBUG_JTDI	n/a	n/a	n/a	
	PB3(JTDO/TRACESWO)	DEBUG_JTDO-SWO	n/a	n/a	n/a	
	PB4(NJTRST)	DEBUG_NJTRST	n/a	n/a	n/a	
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	ACCEL_SCL
	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	ACCEL_SDA
TIM1	PA7	TIM1_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	TXPULSE_A
	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	TXPULSE_B
TIM2	PB10	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM8	PB12	TIM8_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LOOP2_DAC
TIM15	PA3	TIM15_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	LOOP1_DAC
UART4	PB14	UART4_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	RS485_DE
	PA12	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RS485_TX
	PB8	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RS485_RX
USART1	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
Single Mapped Signals	PB2	LPTIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA11	ADC1_EXTI11, ADC2_EXTI11	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	ADC_TRIG_IN
GPIO	PC13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EBLRX
	PC14-	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ACCEL_INT1

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	OSC32_IN(OSC32_IN)					
	PC15-OSC32_OUT(OSC32_O	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	ACCEL_INT2
	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	T_LED
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	nLED
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RS485_RE
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	WIPER_MOTOR
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	nSTATUS_LED

4.2. GPDMA1

4.3. GPDMA2

4.4. LINKEDLIST

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	14	0
RTC non-secure interrupt	true	0	0
RCC non-secure global interrupt	true	0	0
ADC1 global interrupt	true	0	0
TIM1 Break interrupt	true	0	0
TIM1 Update interrupt	true	0	0
TIM1 Trigger and Commutation interrupts	true	0	0
TIM1 Capture Compare interrupt	true	0	0
TIM4 global interrupt	true	0	0
TIM6 global interrupt	true	15	0
I2C1 Event interrupt	true	0	0
I2C1 Error interrupt	true	0	0
USART1 global interrupt	true	0	0
UART4 global interrupt	true	0	0
ADC2 global interrupt	true	0	0
DTS global interrupt	true	0	0
Flash non-secure global interrupt	unused		
EXTI Line11 interrupt	unused		
IWDG global interrupt	unused		
DAC1 interrupt	unused		
TIM2 global interrupt	unused		
TIM5 global interrupt	unused		
TIM7 global interrupt	unused		
TIM8 Break interrupt	unused		
TIM8 Update interrupt	unused		
TIM8 Trigger and Commutation interrupts	unused		
TIM8 Capture Compare interrupt	unused		
TIM15 global interrupt	unused		
FPU global interrupt	unused		
Instruction cache global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
Data cache global interrupt	unused		
TIM12 global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
RTC non-secure interrupt	false	true	true
RCC non-secure global interrupt	false	true	false
ADC1 global interrupt	false	true	true
TIM1 Break interrupt	false	true	true
TIM1 Update interrupt	false	true	true
TIM1 Trigger and Commutation interrupts	false	true	true
TIM1 Capture Compare interrupt	false	true	true
TIM4 global interrupt	false	true	true
TIM6 global interrupt	false	true	true
I2C1 Event interrupt	false	true	true
I2C1 Error interrupt	false	true	true
USART1 global interrupt	false	true	true
UART4 global interrupt	false	true	true
ADC2 global interrupt	false	true	true
DTS global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h5-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf

Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5930-guidelines-for-power-management-on-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5711-getting-started-with-stm32h5-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5872-introduction-to-the-system-architecture-and-performance-in-the-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4894-how-to-use-eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4992-introduction-to-

	secure-firmware-install-sfi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6014-introduction-to-sfsp-versions-for-stm32h5-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5212-how-to-use-stm32-cache-to-optimize-performance-and-power-efficiency-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5593-how-to-use-the-gpdma-for-stm32-mcus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5959-migrating-within-the-stm32h5-mcus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an6246-guidelines-for-estimating-stm32h5-mcus-lifetime-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4657-stm32-

for related Tools & Software	inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpd-sink-application-with-stm32cubemx-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5925-stm32cube-mcu-package-examples-for-stm32h5-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-

& Software	stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6007-getting-started-with-stirot-st-immutable-root-of-trust-for-stm32h5-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0621-stm32h523xx-and-stm32h533xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm01036972.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0264-stm32-cortexm33-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0481-stm32h52333xx-stm32h56263xx-and-stm32h573xx-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf

Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3150-stm32h5-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3267-stm32h5-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf